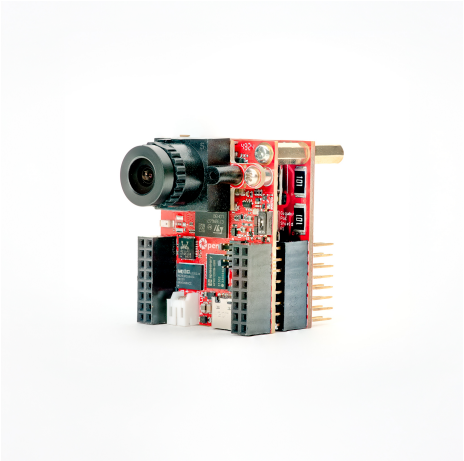
OpenMV® Gigabit PoE Shield

Datasheet

SKU: OMV-SHIELD-GIGABIT-POE

Description

The Gigabit PoE Shield lets you connect your OpenMV Cam to PoE Ethernet with gigabit speeds, providing a one-cable solution for power and connectivity. It features a 10/100/1000 Mb/s capable Ethernet jack with IEEE 802.3af PoE support providing up to ~6 W of power. The shield supplies 5.6 V via VIN through an ideal OR'ing diode, allowing it to be combined with other dual header column shields. Peak UDP transmit performance exceeds 500 Mb/s using Python sockets with the N6.



Target Areas

High-bandwidth vision, video streaming, IP camera, industrial Ethernet, gigabit PoE deployments, high-throughput data transfer.

Key Features

- **Gigabit Ethernet** — 10/100/1000 Mb/s capable Ethernet jack.
- **IEEE 802.3af PoE** — Power over Ethernet providing up to ~6 W to your OpenMV Cam.
- **500+ Mb/s throughput** — Peak UDP transmit performance exceeds 500 Mb/s with N6.
- **One-cable solution** — Single Ethernet cable provides both power and gigabit data.
- **Shield stacking** — Supplies 5.6 V on VIN via ideal OR'ing diode for multi-shield combinations.

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1 Features

1.1 Ethernet

Component	Details
Speed	10/100/1000 Mb/s Gigabit Ethernet.
PoE	IEEE 802.3af compliant. Provides up to ~6 W of power.
Performance	Peak UDP transmit exceeds 500 Mb/s using Python sockets (tested with N6).

1.2 Power Output

Component	Details
VIN	Supplies 5.6 V via VIN through ideal OR'ing diode. Compatible with other dual header column shields.

1.3 Compatibility

Component	Details
Supported Cameras	Only OpenMV Cams with onboard Ethernet: OpenMV N6 and OpenMV Cam RT1062.

Comes with mounting hardware and screws.

2 General Info

Dimension	Value
Weight	23 g

3 Company Information

Field	Details
Website	https://openmv.io

4 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/gigabit-poe-shield
Documentation	https://docs.openmv.io

5 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.